

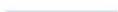
Python for Data Science I

Write clear, testable Python and reason about data transformations.



Learning outcomes

01  Use core Python types and control flow intentionally.

02  Write small functions with explicit inputs and outputs.

03  Debug type, scope, and indexing errors.

The concept system

01

data types

02

control flow

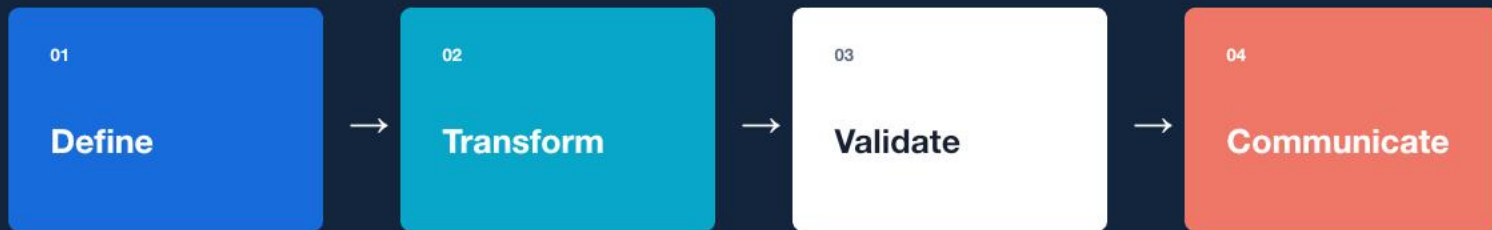
03

functions

04

exceptions

A repeatable reasoning loop



Build a reusable profiler for a list of transaction records.

Guided lab

LAB BRIEF

Build a reusable profiler for a list of transaction records.

workbook / tests / evidence

01

Frame

Define inputs, outputs, and success criteria.

02

Build

Implement the smallest correct workflow.

03

Verify

Test normal, boundary, and failure cases.

04

Explain

Record the evidence and remaining limits.

Portfolio deliverable

A tested Python module and concise debugging log.

QUALITY GATE

-  **Reproducible**
-  **Tested**
-  **Decision-relevant**

WEEKLY ASSESSMENT

Transaction Quality Profiler

Correctness 40%, clarity 25%, tests 20%, explanation 15%.

What should remain after the lesson?

CONCEPT

data types + control flow

PRACTICE

Build a reusable profiler for a list of transaction records.

EVIDENCE

A tested Python module and concise debugging log.
